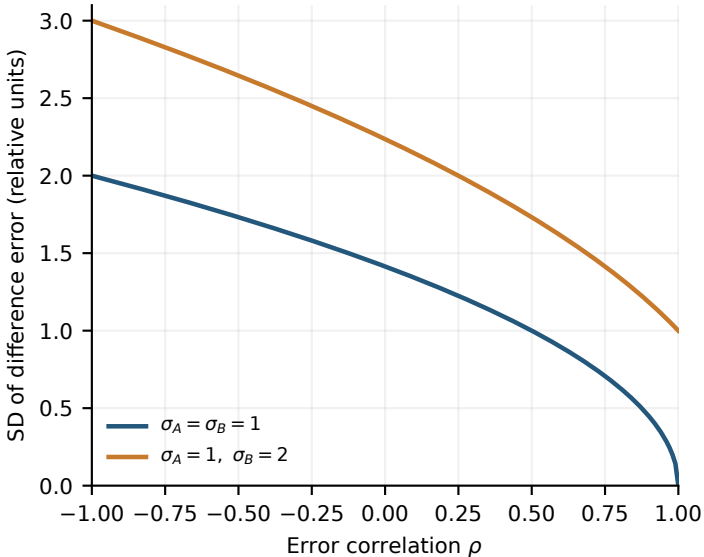
a Correlation changes variance



b Shared-parent measurements

$$d_1 = y_{B_1} - y_A$$

$$d_2 = y_{B_2} - y_A$$

$$\mathrm{Cov}(\eta_{d_1}, \eta_{d_2}) = \mathrm{Var}(\eta_A)$$

For independent assay errors, both differences contain the same parent error. Shared comparisons are not independent.

$$\mathrm{Var}(\hat{\Delta} - \Delta) = \sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B$$

Illustrative error model; variance reduction does not remove bias or establish a reliable direction.